

Aleksei Lian

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EDUCATION

York University

Bachelor of Science in Computer Science

Toronto, ON

Expected Dec. 2027

EXPERIENCE

Software Engineer (Co-Founder)

May 2026 – Present

Arzan (China-to-Kyrgyzstan E-Commerce)

Remote

- Built a 90+ endpoint FastAPI/PostgreSQL API, a React staff console, and a React Native app as the sole engineer.
- Fixed a race where two operators could buy the same Taobao order by adding PostgreSQL row locks, and made checkout idempotent so a retried request never places a second order.
- Enforced money, quantity, and status rules with 50 PostgreSQL CHECK constraints that reject invalid data.
- Added TOTP two-factor sign-in and an audit log of privileged actions for staff accounts.
- Set up GitHub Actions CI that runs the 2,300+ test backend suite on both SQLite and PostgreSQL.

Backend Engineer (Co-Founder)

Apr. 2026 – Aug. 2026

Tanish (Dating App for Central Asia, iOS Beta)

Remote

- Shipped an iOS dating app to a TestFlight beta: FastAPI/PostgreSQL backend, WebSocket chat, React Native client.
- Designed JWT auth with 15-minute access tokens and rotating refresh tokens that revoke their whole chain on reuse.
- Integrated AWS Rekognition face-liveness checks to verify users are real, and GPT-4o-mini scoring of match answers that adjusts compatibility; localized into 4 languages.
- Closed a hole where Supabase's auto-generated API could read and write user tables with the project's public key, by enabling deny-by-default row-level security.

Math Tutor

Jan. 2024 – Present

Self-Employed

Remote

- Tutor students one-on-one in math: pinpoint the exact step where a solution goes wrong and explain it a different way.

PROJECTS

Temir AI - Quote Builder for a Metal Supplier | JavaScript, Anthropic API, PostgreSQL

Aug. 2026 – Sep. 2026

- Built an AI tool, used by **4 sales managers at 2 warehouses**, that turned customer orders sent as chat text, PDF, spreadsheet, or photo into priced quotes.
- Kept the LLM out of quote pricing (it only extracts line items, and every price comes from the catalog) and capped API spend per user and company-wide with a PostgreSQL advisory lock.

Mendy - AI Advice Companion | Python, FastAPI, PostgreSQL, OpenAI API

Jul. 2026

- Built an AI advice companion's backend: long-term memory, proactive follow-ups, and escalation of crisis messages.
- Wrote a 40-message safety test set, re-run after every prompt change, that must flag all 12 crisis cases; it caught a prompt edit that moved a crisis message to a lower risk tier.

URL Shortener | Python, FastAPI, asyncpg, PostgreSQL, Docker, GCP

Feb. 2026 – Mar. 2026

- Built an async URL shortener where two simultaneous requests can't get the same short code: inserts retry on a primary-key conflict instead of check-then-insert.
- Deployed it to Google Cloud Run with Cloud SQL via Cloud Build, with GitHub Actions CI running 38 tests, 23 against a live PostgreSQL container.

Chess Engine | Python, UCI

Feb. 2026

- Built a chess engine that plays full timed games in standard chess GUIs such as Banksia, using alpha-beta search, a transposition table, and time management.

SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Java, C

Frameworks: FastAPI, SQLAlchemy 2.0 (async), Pydantic, Alembic, pytest, React, React Native

Data & Cloud: PostgreSQL (transactions, locking), Redis, SQLite, AWS, GCP (Cloud Run, Cloud SQL), OpenTofu, Supabase

Tools: Git, GitHub Actions, Docker, Linux, REST APIs, WebSockets, Claude Code (AI-assisted development)